Rotaract Club of Xiamen University Malaysia

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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Africoders
Selected Domain	Domain 1: Workforce Domain
Selected SDG Track (1–17)	8. DECENT WORK AND ECONOMIC GROWTH
Project / Solution Name	CareerAtlas AI
Name (Leader)	Azza Ladhgham
Email & Phone (Leader)	+60 17-206 7172 : cst2409090@xmu.edu.my
University Affiliation(s)	Xiamen University Malaysia

Number of Team Members 4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

The world is witnessing the rapid evolution of AI and how it is reshaping the job landscape, transforming roles and giving birth to new ones. Students and graduates, professionals and employers are increasingly uncertain about the emerging and disappearing careers, and the skills required to stay competitive.

One of the key issues is the regional diversity of the effects of AI on jobs. Careers that are shrinking in one country could be expanding in another, as a result of varying rates of tech adoption, economic growth and industry needs. These variations are frequently not taken into account in the generalized forecasts that are made globally, which in turn restricts the awareness of opportunities in their home or other countries.

This information gap rises skills mismatches, risk of unemployment and economic inequalities, and leads to the loss of valuable opportunities within a locality's economy, making it less attractive to others, resulting in unnecessary talent leakage and potentially reducing local workforce development.

We are suggesting an AI driven Future Workforce Intelligence Platform that will be able to fuse Geographical Intelligence to uncover new and vanishing careers and skills by country and continent. It will offer users a glimpse into the local and international job market as well as customized recommendations for jobs and relevant reskilling courses.

Our solution’s not only about helping individuals choose their career paths but also about fostering local talent development, building strong national workforces and promoting economic growth through increased awareness and access to domestic opportunities.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

We propose an AI enabled Future Workforce Intelligence Platform for students, graduates, professionals and employers to help them navigate the evolving job market, explore future career paths and prepare for the impact of AI on jobs.

The platform will leverage AI and Generative AI (GenAI) to process regularly updated labor-market data, job postings, and technological advancements to determine both emerging and declining jobs and skills. By pinpointing the geographical distribution of these changes across countries and continents, our central innovation, Geographical Intelligence, will expand this analysis. This will help users to compare the workforce situation in their region and identify new job prospects in their own country and abroad, and to make informed career decisions. The platform will also help raise awareness of the opportunities in the local community, promote local talent development, and support the national economy.

Users will create skills profiles through CV uploads or and filling in their skills. The system will align these profiles with the AI workforce trends and offer relevant future career suggestions, explainable matching scores and skills gaps. It will then create customized reskilling trajectories, with estimated learning times and available learning resources. Moreover, the platform will also examine several scenarios of AI disruption to address uncertainty and link suggested careers and skills to the Sustainable Development Goals (SDGs) of the United Nations.

We will use technical expertise gained from the SDC Phase 1 workshops to implement the solution, such as GenAI APIs for interpreting trends and making tailored recommendations; database architecture to store skills profiles and labor-market information; and cloud hosting to ensure our web application is accessible. These technologies will enable an evidence based, scalable platform for lifelong learning, workforce adaptability and sustainable economic development.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

CareerAtlas AI is an AI-driven workforce intelligence tool that supports students, graduates and professionals throughout an evolving job market with informed career decisions. It tackles career insecurity, lack of skills match and difficulty accessing geographically relevant workforce information as a result of its AI, occupational data and tailored career recommendations.

It features a React and TypeScript user interface, an Express.js back end, and well-organized JSON datasets. It has a React and TypeScript UI, an Express.js back end, and well-structured JSON datasets. It combines information on occupational and skills with information on employment from ILOSTAT and the statistics agency of Malaysia (DOSM), generative AI exposure data from the International Labour Organization (ILO), and signals of workforce trends from the arXiv and Hacker News Algolia APIs.

CareerAtlas AI uses an explainable matching algorithm that calculates both existing skill score and how well it matches the skills needed for a job. It takes into account the skill momentum of the user and exposure to AI. It matches relevant career opportunities, pinpoints skill deficits, and suggests individual reskilling routes.

It has Geographical Intelligence which provides a workforce overview relevant to the country that the user chooses.

The platform is designed to integrate Claude's generative AI features to extract relevant skills from CV text and offer explanations on career recommendations. Unlike determining career ranking, fuse.js allows for approximate skill matching and Claude allows for generating understandable explanations with structured results.

The application uses modules that have been built in the SDC Phase 1 workshops, such as integrating with AI APIs, structuring the data, and deploying it in the cloud, among others. The functional prototype is deployed using Replit to access CareerAtlas AI via a web browser.

Yes. We made use of a cloud-based development and deployment platform called Replit which was offered as free credits during the SDC Hackathon. These credits enabled the functional web prototype of the CareerAtlas AI to be deployed, so that users can access and interact with the application online. The application is hosted and tested from the cloud in Replit, and additionally Claude is integrated separately to assist with explanations for the AI-powered skill extraction and career recommendations.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We used a user-centered design-thinking approach to create CareerAtlas AI as a technically viable, accessible and practical workforce intelligence platform.

Empathized: We found students, graduates, and professionals are uncertain about the future of emerging careers, hard to determine skill gaps, and lack access to information on workforce in their areas.

Define: We identified core problem as personalized, explainable, geographically relevant career guidance in the rapidly changing workforce.

Ideate: Discussed and identified potential solutions and chose a combined platform that integrates AI-driven career matching, workforce trend intelligence, geographical data, skill gap analysis and personalized reskilling advice.

Since we were only four people in our team, we split the development tasks in the following way: data engineering, development of matching engine and API, frontend development, documentation. We developed our application with React, TypeScript, Express.js, Claude, and organized datasets. Collaboration was made possible with GitHub, and cloud deployment was achieved with Replit.

To assess the career recommendations without releasing real personal information, we created synthetic user profiles for testing. Data validation scripts, dated workforce snapshots, source attribution, and explanations of the scoring are included in the system to boost reliability. Testing also takes into account scenario-based recommendations and usability of the applications.

The CareerAtlas AI is deployed on Replit as a working prototype of a web application. Its modular design allows for the separation of the user interface, matching engine, AI services, and data-processing pipeline, enabling easy maintenance and future expansion. To update a workforce trend, use reviewable data snapshots to automate the process with GitHub Actions.

In Phase 2, we want to add interactivity to geographical maps, add cross-country career comparisons, enrich the visualizations of the workforce, extend the range of countries covered, and improve personalized recommendations.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4.4 — Skills for Employment Supports relevant technical and vocational skills development through personalized reskilling pathways. SDG 8.5 — Decent Work and Employment Helps people prepare for changing employment opportunities and identify suitable future careers. SDG 10.2 — Social and Economic Inclusion Aims to make career intelligence more accessible, including information about opportunities within users' own countries
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?*	AI exposure decreases by country income level, with high exposure for just 12 percent of workers in low-income

Metric Category	Your Response
e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."	countries and 15 percent of workers in lower-middle-income countries. These results suggest that for developing countries, and in particular low-income countries, the labor market impacts of artificial intelligence will be more limited than in high-income countries. Demombynes, G., Langbein, J., & Weber, M. (2025). *The exposure of workers to artificial intelligence in low- and middle-income countries* (Policy Research Working Paper No. 11057). World Bank.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage (%) of users who successfully identify at least one suitable, emerging career opportunity within their country and receive a personalized reskilling pathway aligned with its skill requirements.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 80% of test users will successfully identify one suitable, emerging career opportunity within their country and receive a personalized reskilling pathway aligned with its required skills using our platform by the final prototype presentation.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ no

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ no

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Code	MIT
Vite	Code	MIT
Tailwind CSS	Code	MIT
Recharts	Code	MIT

Express.js	Code	MIT
Fuse.js	Code	Apache 2.0
Claude (Anthropic)	AI Model	Anthropic Commercial Terms
ESCO	Dataset	CC BY 4.0
ISCO-08 (ILO)	Datase/classification	ILO usage terms
ILO AI Exposure Data	Dataset	ILO publication/data reuse terms
ILOSTAT	Dataset	ILOSTAT data usage terms
DOSM Employment Data	Dataset	DOSM data usage terms
arXiv API	Data API	arXiv API terms
Hacker News Algolia API	Data API	API/provider usage terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

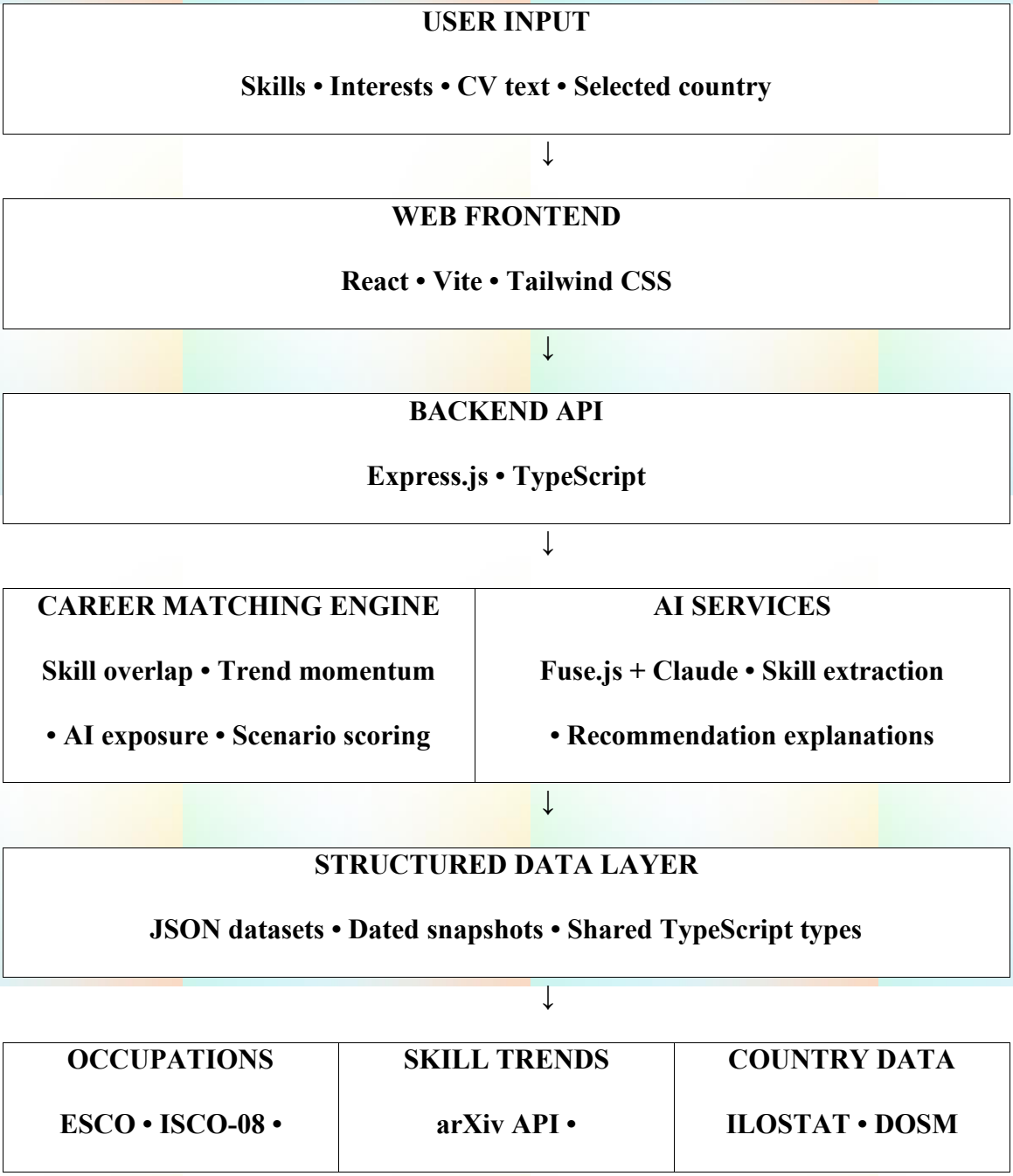
[Signature]



[Date]

24/10/2026

Appendix A: CareerAtlas AI System Architecture



ILO AI exposure	**Hacker News Algolia**	

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USER RESULTS **Career recommendations • Skill gaps • Reskilling pathways • Geographical insights • SDG alignment**

CareerAtlas AI is architectured with modules that link the React front-end to an Express.js back-end. The back end is made up of an explainable Career Matching engine, Claude AI services, and structured data on the workforce. Occupational and workforce information is sourced from external data such as ESCO, ILO, arXiv, Hacker News, ILOSTAT and DOSM. Data can be automatically updated with GitHub Actions. The platform analyzes user profiles to provide career recommendations, skill gaps, personalized learning paths, and location-specific workforce insights. The functional prototype is deployed via Replit.